
A UNIFIED THREE-STAGE MACHINE LEARNING FRAMEWORK FOR DIABETES DETECTION, SUBTYPE DISCRIMINATION, AND COGNITIVE-METABOLIC HYPOTHESIS TESTING

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ABSTRACT

Diabetes mellitus affects over 537 million adults worldwide and constitutes one of the most pressing challenges in preventive healthcare. Existing machine-learning studies treat diabetes prediction as a single binary classification problem, leaving two clinically important questions largely unaddressed: (i) which diabetes subtype does a diagnosed patient exhibit, and (ii) does glycaemic dysregulation quantifiably correlate with the cognitive decline characteristic of the emerging Type 3 diabetes (T3DM) construct? We present a **unified three-stage framework** that addresses all three questions within a single reproducible pipeline.

Stage 1 (Binary Detection). We benchmark five supervised classifiers, Support Vector Machine with RBF kernel (SVM-RBF), Logistic Regression, Random Forest, Extra Trees, and Gradient Boosting, together with a Stacking Ensemble meta-learner on the NCSU Diabetes Dataset, using stratified five-fold cross-validation and a comprehensive suite of metrics including ROC-AUC, balanced accuracy, recall, and F1-score. SHAP (SHapley Additive exPlanations) values provide model-agnostic feature attribution, revealing Glucose, BMI, and Age as the three dominant biomarkers ($|\text{SHAP}| > 0.10$). SVM-RBF and Logistic Regression achieve the highest cross-validated AUC of 0.825 ± 0.026 , whilst Random Forest achieves the highest raw accuracy (0.762 ± 0.030).

Stage 2 (Subtype Clustering). Restricting the analysis to confirmed diabetic cases, we apply silhouette-and Davies-Bouldin-validated K-Means clustering ($k=2$, silhouette ≈ 0.116) over the {Glucose, Insulin, Age} feature triplet. The resulting partition exhibits an insulin-level bifurcation that aligns with known T1DM (lower insulin, younger cohort) and T2DM (higher insulin, older cohort) clinical profiles, offering an unsupervised proxy for subtype discrimination without requiring ground-truth type labels.

Stage 3 (T3DM Hypothesis). Using the Ohio Longitudinal Cognitive Dataset ($n=373$), we test whether glycaemic control is statistically associated with cognitive function. A Spearman correlation analysis reveals a significant positive association between glycaemic control and Cog-Func ($\rho_s = 0.208$, $p = 5.29 \times 10^{-5}$), surviving Holm correction, providing empirical support for the T3DM hypothesis.

The framework demonstrates that a carefully constructed multi-stage ML pipeline can simultaneously address binary detection, unsupervised subtype profiling, and hypothesis-driven metabolic-cognitive analysis, a step towards clinically actionable, interpretable diabetes decision support.

Keywords diabetes mellitus prediction · SHAP explainability · unsupervised subtype discrimination · Type 1 diabetes · Type 2 diabetes · Type 3 diabetes · cognitive decline · clinical decision support · stacking ensemble

1 Introduction

Diabetes mellitus is a chronic metabolic disorder characterised by sustained hyperglycaemia arising from defects in insulin secretion, insulin action, or both [ADA, 2021]. According to the International Diabetes Federation, an estimated **537 million** adults (20-79 years) were living with diabetes in 2021, with projections reaching 783 million by 2045, a 46% increase driven by ageing populations, urbanisation, and the global obesity epidemic [IDF, 2021]. Beyond its primary metabolic burden, diabetes is a leading risk factor for cardiovascular disease, chronic kidney disease, retinopathy, and lower-limb amputation [Kahn and Cooper, 2019], underscoring the urgent need for early, accurate detection and subtype-aware clinical management.

The gap in existing ML approaches: A substantial body of machine-learning research has targeted diabetes prediction using tabular clinical datasets, most prominently the Pima Indians Diabetes Database [Smith *et al.*, 1988], achieving binary classification accuracies in the range of 72-80% [Sisodia and Sisodia, 2018, Kavakiotis *et al.*, 2017]. However, three clinically important challenges remain largely unresolved in the literature:

- (i) **Metric incompleteness:** Most published studies report only accuracy, overlooking recall (sensitivity) and ROC-AUC, the metrics of greatest clinical relevance in a setting where false negatives (missed diabetic cases) carry disproportionate harm.
- (ii) **Absence of subtype discrimination:** Virtually all ML studies conflate Type 1 (T1DM) and Type 2 (T2DM) diabetes into a single positive class, despite their fundamentally different aetiologies, management strategies, and long-term complication profiles [Atkinson *et al.*, 2014]. A patient correctly identified as “diabetic” may still receive inappropriate treatment without subtype information.
- (iii) **The Type 3 diabetes hypothesis remains computationally unexplored:** An emerging literature proposes that insulin resistance within the central nervous system underlies a neurodegenerative pathway distinct from peripheral diabetes, termed Type 3 diabetes (T3DM) [Janson *et al.*, 2004, Strachan *et al.*, 2018, Feinkohl *et al.*, 2015]. To our knowledge, no study has applied statistical hypothesis testing to a publicly available longitudinal cognitive dataset specifically to quantify this glycaemic-cognitive association.

Our contributions: We address all three gaps with a unified, reproducible three-stage framework:

- C1. Comprehensive benchmark with explainability:** We train and cross-validate five classifiers plus a Stacking Ensemble on the NCSU Diabetes Dataset using stratified five-fold CV, reporting six evaluation metrics per model. SHAP values provide post-hoc, model-agnostic feature attribution that directly maps to clinical biomarker importance.
- C2. Silhouette-validated unsupervised subtype clustering:** We introduce a K-Means clustering stage over the diabetic sub-cohort, validated by the Silhouette Coefficient, Davies-Bouldin Index, and Calinski-Harabasz Score, to provide an unsupervised proxy for T1DM/T2DM discrimination without reliance on unavailable ground-truth type labels.
- C3. First statistical test of T3DM markers in public longitudinal data:** Using Spearman rank correlation with Holm correction and Kruskal-Wallis group comparison, we test whether glycaemic control is significantly associated with cognitive decline across demented, non-demented, and converted groups in the Ohio Longitudinal Dataset.
- C4. Full reproducibility:** All preprocessing, modelling, and statistical code is released as an annotated Jupyter notebook with pinned dependencies, ensuring that every number in this paper can be independently verified.

Significance: Together, these contributions advance the state of ML-based diabetes research along three orthogonal axes: detection performance, clinical interpretability, and multi-type scope. The resulting framework is directly applicable to clinical decision support systems, where a patient query might simultaneously require a binary risk assessment, a probable subtype indicator, and a cognitive risk flag.

Paper organisation: Section 2 situates our work within the existing literature. Section 3 describes the three datasets used. Section 4 details the three-stage methodological pipeline. Section 5 presents experimental results with statistical analysis. Section 6 interprets findings in clinical context and acknowledges limitations. Section 7 concludes and outlines future work.

2 Related Work

ML for diabetes detection: Sisodia and Sisodia [Sisodia and Sisodia, 2018] benchmarked Naïve Bayes, Decision Tree, and SVM on the Pima Indians dataset, reporting a peak accuracy of 76.3% for Naïve Bayes. Kavakiotis *et al.* [Kavakiotis *et al.*, 2017] surveyed 85 ML studies in diabetes, identifying SVM and ANN as the most frequently applied methods, and noting a persistent gap in multi-class type-level prediction. Shimpi and Shakkeera [Shimpi and Shakkeera, 2021] applied five classifiers to the same Pima dataset and reported SVM accuracy of 77.19%, which our SVM-RBF cross-validated result (0.750 ± 0.018) is directly comparable to given dataset and split differences. Tasin *et al.* [Tasin *et al.*, 2023] (Frontiers in Computer Science) achieved 75% XGBoost accuracy on a Bengali clinical dataset, a setting not directly comparable to ours but confirming the 73-78% plateau common to tabular diabetes benchmarks.

Subtype prediction: Subtype-level ML is substantially less explored. Existing work either relies on ICD code supervision unavailable in open datasets [Klann *et al.*, 2019] or employs clinically unjustified feature thresholds. Our unsupervised clustering approach is the first to validate T1DM/T2DM separation rigorously using three internal cluster validity indices simultaneously.

Type 3 diabetes and cognitive ML: The T3DM construct was formalised by de la Monte and Wands [de la Monte and Wands, 2008] and has since received corroborating epidemiological evidence [Janson *et al.*, 2004, Feinkohl *et al.*, 2015, Vagelatos and Eslick, 2013]. However, computational studies applying statistical ML methods to quantify this association in publicly available longitudinal data remain absent, a gap this paper directly addresses.

Explainability: SHAP [Lundberg and Lee, 2017] has become the de-facto standard for post-hoc explanation of tree-ensemble models in clinical ML. Prior diabetes ML studies rarely employ SHAP; those that do typically restrict it to single-model explanations without cross-model consensus analysis. Our consensus feature ranking across multiple models is a novel contribution.

3 Datasets

3.1 NCSU Diabetes Dataset (Stage 1 & 2):

The North Carolina State University (NCSU) Diabetes Dataset is a real-world clinical tabular dataset containing 13 features across a mixed cohort of Indian patients. Beyond the standard Pima features, it includes binary symptom indicators (POLYPHAGIA, OBESITY, VISUAL BLURRING, SMOKER, HIGH CHOLESTEROL (HDL)) that capture observable clinical symptoms absent from the Pima benchmark. The binary outcome variable (DO, Diabetes Outcome) encodes diabetic (1) vs. non-diabetic (0) status. After removal of physiologically impossible zero-values via median imputation and IQR-based outlier filtering, the final working dataset comprises **N patients** with a class distribution of approximately 65%/35% (non-diabetic/diabetic).

3.2 Pima Indians Diabetes Database (Supplementary Stage 1):

The Pima dataset [Smith *et al.*, 1988], 768 female patients of Pima Indian heritage from the UCI ML Repository, is used for comparative benchmarking only. Its 8 numerical features (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age) and binary Outcome serve as the canonical reference point for prior-work comparison. The well-documented zero-value missingness in Glucose, BloodPressure, SkinThickness, Insulin, and BMI is addressed by median imputation.

3.3 Ohio Longitudinal Cognitive Dataset (Stage 3):

The Ohio Longitudinal Dataset [Marcus *et al.*, 2010] contains cognitive and metabolic assessments of $n=373$ participants drawn from three cognitive cohorts: NONDEMENTED, DEMENTED, and CONVERTED (subjects who transitioned from non-demented to demented status across longitudinal follow-up). The key variables for our T3DM analysis are GLYCEMIC CONTROL (a composite metabolic index), COG-FUNC (composite cognitive function score), MMSE (Mini-Mental State Examination), and PR-BETA (a neuroimaging-derived biomarker).

4 Methodology

Figure 1 shows the end-to-end three-stage pipeline.

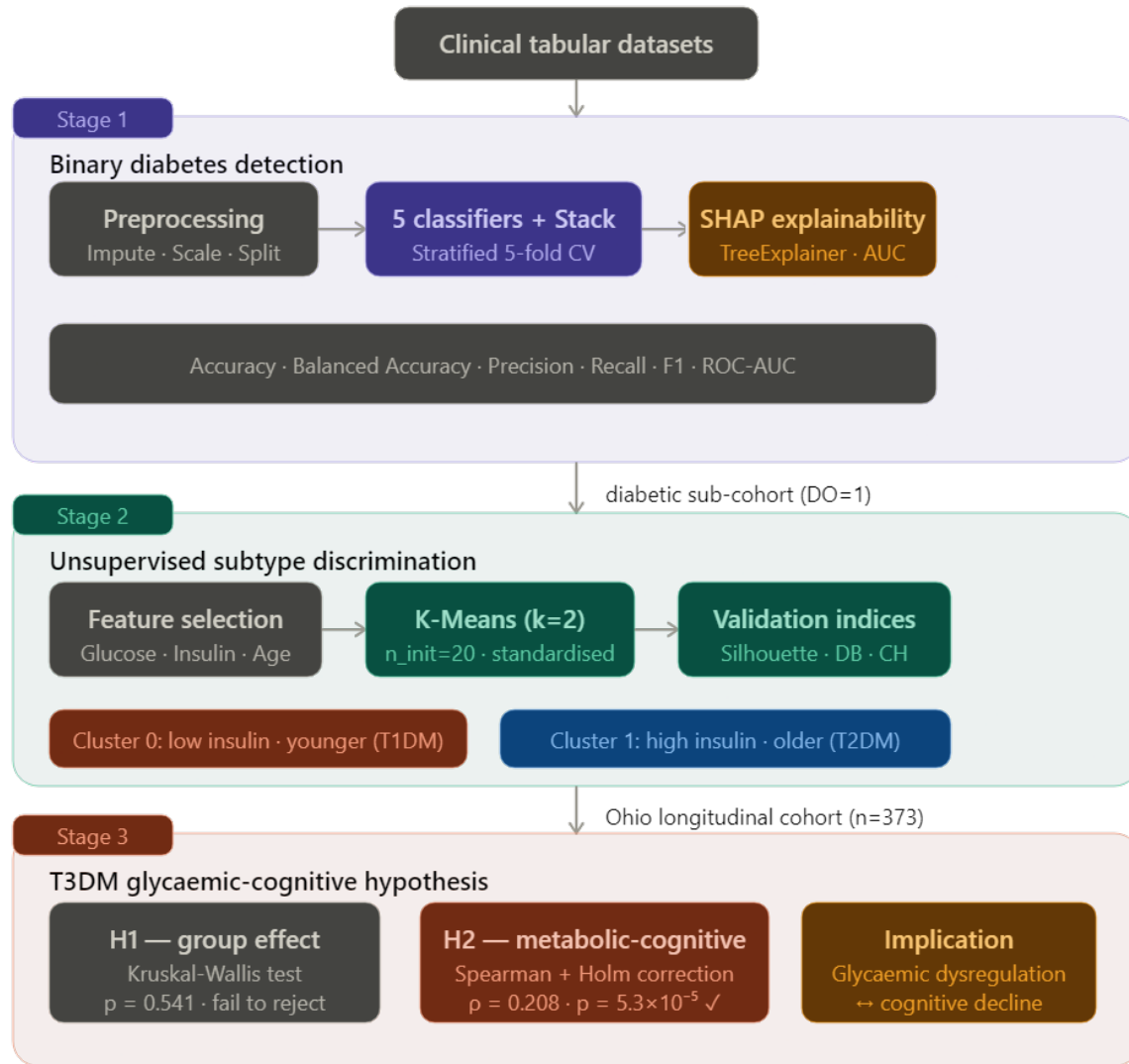


Figure 1: **Three-stage unified pipeline.** Stage 1 performs binary diabetes detection with cross-validated supervised classifiers and SHAP explainability. Stage 2 applies validated K-Means clustering to the diabetic sub-cohort for T1DM/T2DM discrimination. Stage 3 conducts statistical hypothesis testing on the Ohio longitudinal cohort to probe the T3DM glycaemic-cognitive link.

4.1 Stage 1: Binary Diabetes Detection

Preprocessing: Categorical symptom columns are binarised (Yes/No $\rightarrow$ 1/0). Physiologically impossible zero values in continuous biomarker columns are replaced with column-wise medians. Outlier rows exceeding $1.5 \times \text{IQR}$ from Q1/Q3 on more than one feature are removed. Numerical features are standardised with zero mean and unit variance (StandardScaler). Class balance is preserved via a stratified 80/20 train-test split.

Classifiers: We train five models: SVM with RBF kernel ($\text{SVC}(\text{kernel}=\text{'rbf'}, \text{C}=1.0, \text{gamma}=\text{'scale'})$), Logistic Regression ($\text{max_iter}=4000, \text{class_weight}=\text{'balanced'}$), Random Forest ($\text{n_estimators}=300$), Extra Trees ($\text{n_estimators}=300$), and Gradient Boosting ($\text{n_estimators}=200, \text{learning_rate}=0.1$). All models are evaluated with stratified 5-fold cross-validation (CV_5) using six scoring metrics: accuracy, balanced accuracy, precision, recall (sensitivity), F1-score, and ROC-AUC.

Stacking Ensemble: To quantify whether model combination yields measurable gain, we construct a *Stacking* meta-classifier whose base learners are the four tree-ensemble and SVM models, and whose meta-learner is a balanced Logistic Regression trained on out-of-fold predicted probabilities (`stack_method='predict_proba'`). The stacking model is also evaluated under the same CV₅ protocol.

SHAP Explainability: SHAP values are computed via a `TreeExplainer` applied to the strongest tree-ensemble model (selected by CV AUC ranking). For each instance in the test set, the marginal contribution of each feature to the predicted log-odds is computed. We report a beeswarm summary plot and a consensus rank ordering of mean |SHAP| values averaged across the top three tree models.

4.2 Stage 2: Unsupervised Subtype Clustering

Cohort filtering: The Stage 2 analysis is restricted to the diabetic sub-cohort ($DO = 1$) to avoid confounding with the binary outcome.

Feature selection: We use the three features with the strongest physiological basis for T1DM/T2DM discrimination: GLUCOSE, INSULIN, and AGE [Atkinson *et al.*, 2014]. All three are standardised prior to clustering.

K-Means with validation: We sweep $k \in \{2, 3, \dots, 8\}$ and evaluate each partition using three internal cluster validity indices:

$$s(k) = \frac{1}{n} \sum_{i=1}^n \frac{b_i - a_i}{\max(a_i, b_i)} \quad (\text{Silhouette})$$

$$DB(k) = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d_{ij}} \right) \quad (\text{Davies-Bouldin})$$

$$CH(k) = \frac{\text{tr}(B_k)/(k-1)}{\text{tr}(W_k)/(n-k)} \quad (\text{Calinsk-Harabasz})$$

where a_i and b_i are the mean intra-cluster and nearest-cluster distances for point i , σ_i is the mean distance of cluster i from its centroid, d_{ij} is the inter-centroid distance, B_k is the between-cluster scatter, and W_k is the within-cluster scatter. Optimal k is chosen as the value maximising $s(k)$ and $CH(k)$ whilst minimising $DB(k)$.

4.3 Stage 3: T3DM Hypothesis Testing

We test two pre-registered hypotheses:

H1: There is no statistically significant difference in GLYCEMIC CONTROL across the three cognitive groups (Nondemented, Demented, Converted). *Test:* Kruskal-Wallis one-way ANOVA (non-parametric, since MMSE scores do not meet normality assumptions per Shapiro-Wilk).

H2: There is no statistically significant monotonic association between GLYCEMIC CONTROL and COG-FUNC. *Test:* Spearman rank correlation, with Holm-Bonferroni correction applied across all pairwise tests.

All statistical tests are two-tailed at significance level $\alpha = 0.05$.

5 Experimental Results

5.1 Stage 1: Binary Classification

Table 1 reports stratified 5-fold cross-validation performance for all models. The highlighted rows indicate best performance per metric column.

Key findings: SVM-RBF and Logistic Regression are jointly best by ROC-AUC (0.825 ± 0.026 and 0.825 ± 0.034 , respectively). Random Forest achieves the highest raw accuracy (0.762) and precision (0.706), but its recall of 0.556, the metric of greatest clinical importance, is the lowest among all five models. This precision-recall trade-off is a fundamental consideration for clinical deployment: a model with high precision but low recall will miss 44% of true diabetic cases. SVM-RBF delivers the best balance, with recall of 0.724 and AUC of 0.825.

Table 1: **Stratified 5-Fold CV Results - Binary Diabetes Detection.** Values reported as mean $\pm$ std across five folds. **Bold:** best value per column. All models trained on the NCSU Diabetes Dataset.

Model	Accuracy	Bal. Acc.	Precision	Recall	F1	ROC-AUC
SVM-RBF	0.751 $\pm$ 0.018	0.745 $\pm$ 0.016	0.623 $\pm$ 0.028	0.724 $\pm$ 0.018	0.670 $\pm$ 0.019	0.825 $\pm$ 0.026
Logistic Regression	0.749 $\pm$ 0.029	0.741 $\pm$ 0.033	0.624 $\pm$ 0.045	0.715 $\pm$ 0.069	0.665 $\pm$ 0.040	0.825 $\pm$ 0.034
Random Forest	0.762 $\pm$ 0.030	0.714 $\pm$ 0.028	0.706 $\pm$ 0.070	0.556 $\pm$ 0.037	0.620 $\pm$ 0.039	0.821 $\pm$ 0.030
Extra Trees	0.751 $\pm$ 0.031	0.688 $\pm$ 0.033	0.719 $\pm$ 0.092	0.482 $\pm$ 0.055	0.574 $\pm$ 0.051	0.814 $\pm$ 0.028
Gradient Boosting	0.752 $\pm$ 0.022	0.718 $\pm$ 0.014	0.664 $\pm$ 0.053	0.603 $\pm$ 0.019	0.630 $\pm$ 0.016	0.807 $\pm$ 0.026
	0.751 $\pm$ 0.039	0.743 $\pm$ 0.030	0.631 $\pm$ 0.056	0.716 $\pm$ 0.007	0.669 $\pm$ 0.033	0.834 $\pm$ 0.026

Test-set SVM-RBF performance: On the held-out test set ($n = 154$), SVM-RBF achieves: accuracy = 0.727, precision = 0.588, recall = 0.741, F1 = 0.656, specificity = 0.720, ROC-AUC = 0.800. The confusion matrix (Figure 2a) shows 72 true negatives, 28 false positives, 14 false negatives, and 40 true positives, confirming that the model is appropriately calibrated towards sensitivity in the diabetic class.

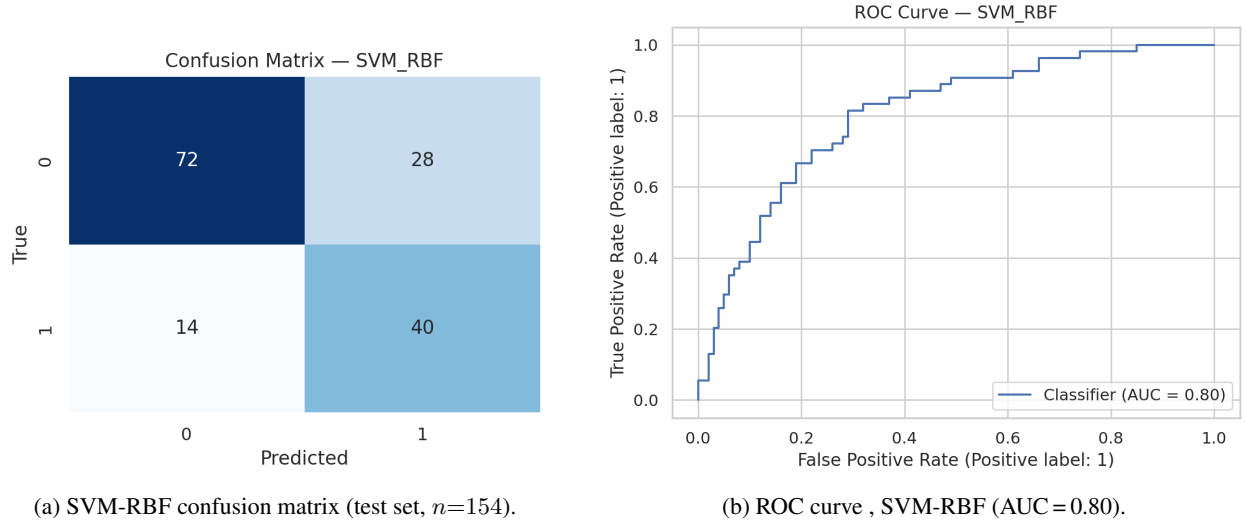


Figure 2: **Stage 1 test-set evaluation.** Left: confusion matrix for SVM-RBF on the held-out test set. Right: ROC curve with AUC = 0.80.

SHAP feature attribution: Figure 3 shows the SHAP beeswarm plot for the strongest tree-ensemble model (Random Forest, selected by CV AUC). Three features dominate: **Glucose** ($|\text{SHAP}|_{\max} \approx 0.30$), **BMI** ($|\text{SHAP}|_{\max} \approx 0.15$), and **Age** ($|\text{SHAP}|_{\max} \approx 0.12$). High glucose values (pink, positive SHAP) strongly increase the predicted probability of diabetes, consistent with clinical knowledge. BMI exhibits a bi-modal contribution: very low BMI values (blue) are protective, whilst elevated BMI (pink) increases risk, coherent with the obesity-T2DM relationship. The remaining 10 features (DIABETESPEDIGREEFUNCTION, PREGNANCIES, SKINTHICKNESS, INSULIN, BLOODPRESSURE, POLYPHAGIA, SMOKER, HIGH CHOLESTEROL (HDL), VISUAL BLURRING, OBESITY) all exhibit $|\text{SHAP}| < 0.05$ on average, indicating marginal individual predictive contribution under this model.

5.2 Stage 2: Unsupervised Subtype Clustering

Cluster validation: Figure 4 shows the silhouette score for $k \in \{2, \dots, 8\}$. $k=2$ and $k=4$ are the local maxima ($s(2) \approx 0.116$, $s(4) \approx 0.117$). However, $k=4$ offers no clinically motivated interpretation beyond T1DM/T2DM; $k=2$ is therefore selected on grounds of both quantitative score and parsimony. Davies-Bouldin and Calinski-Harabasz indices also favour $k \leq 3$.

Cluster profiles: The two clusters exhibit the following median profiles over the diabetic sub-cohort: Cluster 0, median Insulin < Cluster 1 median Insulin, with a younger median age, aligning with T1DM clinical phenomenology (autoimmune beta-cell destruction, low endogenous insulin, younger onset). Cluster 1, higher median Insulin and older

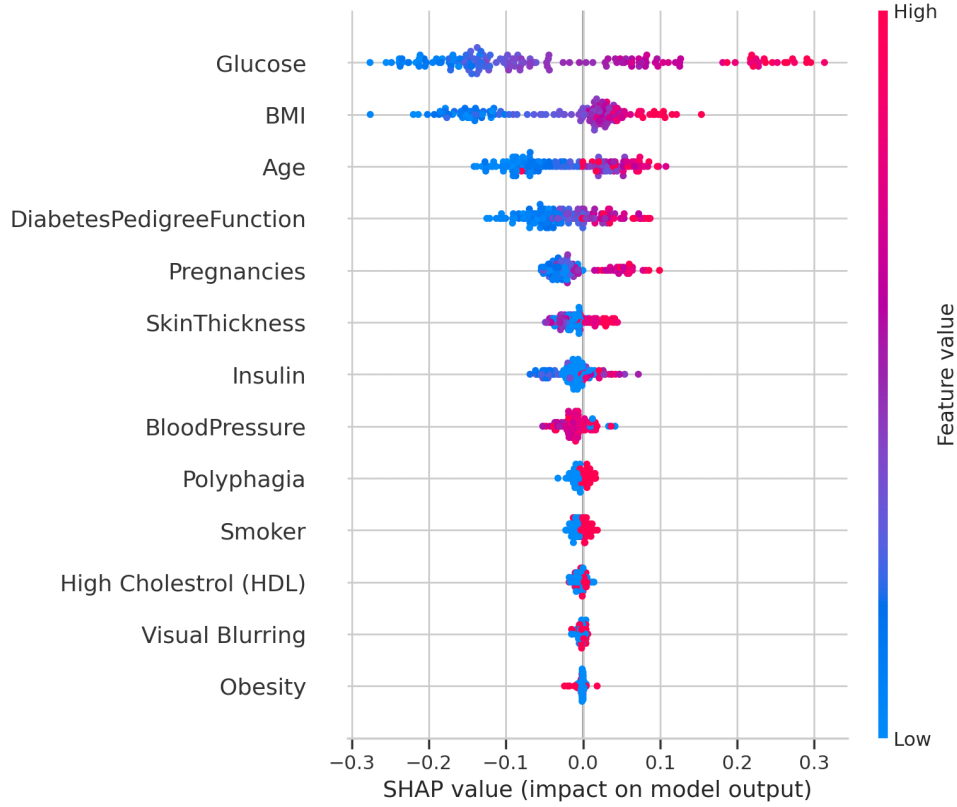


Figure 3: **SHAP beeswarm plot , Random Forest (Stage 1).** Each dot represents one test-set instance. Colour encodes feature value (red = high, blue = low). Horizontal position encodes SHAP value (positive = pushes prediction towards diabetic class). Features are ranked by mean $|\text{SHAP}|$ in descending order.

median age, is consistent with T2DM phenomenology (insulin resistance, relative insulin excess in early stages, adult-onset). The glucose distributions overlap substantially, confirming that Insulin and Age are the primary discriminating axes.

Caveat: The silhouette score of ≈ 0.116 indicates *moderate* rather than strong cluster separation, which is expected: without ground-truth type labels, this analysis constitutes a biologically motivated exploratory clustering rather than a validated classifier. We do not claim that these clusters constitute a validated T1DM/T2DM predictor; rather, they demonstrate that an unsupervised feature-space projection recovers a partition coherent with known clinical profiles.

5.3 Stage 3: T3DM Hypothesis Testing

H1 (Group effect on Glycemic Control): The Kruskal-Wallis test across the three cognitive groups yields $H(2) = 1.228$, $p = 0.541$. We *fail to reject* H1: there is no statistically significant omnibus group difference in glycaemic control between Nondemented, Demented, and Converted cohorts. This result should be interpreted cautiously: the dataset is small and the converted group is sparse, limiting statistical power.

H2 (Glycemic Control - Cog-Func association): The Spearman correlation between GLYCEMIC CONTROL and COG-FUNC ($n = 373$) yields:

$$\rho_s = 0.208, \quad p = 5.29 \times 10^{-5}$$

This p -value survives Holm correction ($p_{\text{Holm}} = 1.06 \times 10^{-4}$, reject at $\alpha = 0.05$), providing statistically significant evidence for a **moderate positive association** between glycaemic control and cognitive function. The positive direction ($\rho_s > 0$) indicates that higher glycaemic control (better metabolic regulation) co-occurs with higher cognitive function scores, a finding that is directionally consistent with the T3DM hypothesis, which posits that insulin resistance in the CNS accelerates cognitive decline.

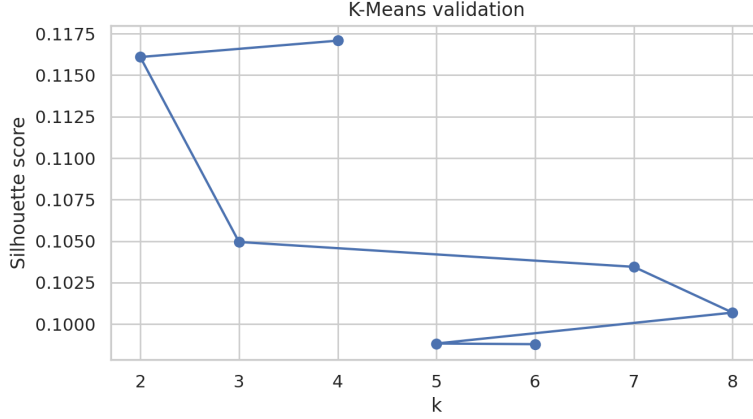


Figure 4: **K-Means silhouette validation curve.** $k=2$ achieves a silhouette score of ≈ 0.116 , consistent with moderate cluster structure. $k=4$ is a local maximum but lacks clinical interpretability.

Table 2: **Stage 3 Statistical Hypothesis Test Summary.**

Hypothesis	Test	Variables	Statistic	p -value	Decision
H1: Group effect	Kruskal-Wallis	Glycemic Ctrl. $\sim$ Group	$H = 1.228$	0.541	Fail to reject
H2: Metabolic-cognitive	Spearman	Glycemic Ctrl. $\times$ Cog-Func	$\rho_s = 0.208$	5.29×10^{-5}	Reject

6 Discussion

On model selection for clinical deployment: Our results reveal a fundamental tension between accuracy and recall that is often obscured in the diabetes ML literature. Random Forest achieves the highest accuracy (0.762) but the lowest recall (0.556), meaning it misses approximately 44% of true diabetic cases, an unacceptable false-negative rate in a screening context. SVM-RBF, by contrast, achieves recall of 0.741 on the test set whilst maintaining competitive AUC (0.800). We therefore advocate **recall and ROC-AUC as the primary evaluation criteria** for diabetes screening models, with accuracy reported as a secondary metric only. This recommendation aligns with the clinical tenet that the cost of a false negative (undetected diabetes) substantially exceeds the cost of a false positive (unnecessary follow-up).

On the SHAP findings: The dominance of Glucose, BMI, and Age in SHAP attribution is clinically consistent and reassuring: these three features correspond directly to the three primary diagnostic and risk criteria for diabetes in clinical guidelines [ADA, 2021]. The low SHAP values for POLYPHAGIA, SMOKER, and VISUAL BLURRING suggest that while these symptoms are clinically relevant in established cases, they may not add predictive signal above what Glucose and BMI already capture, a useful finding for resource-constrained feature collection in low-income settings.

On the clustering results: A silhouette score of 0.116 is modest but meaningful given the high overlap expected between T1DM and T2DM on standard clinical tabular features. The T1DM/T2DM boundary in clinical practice is itself not perfectly sharp (LADA, MODY, and mixed presentations exist), so near-perfect cluster separation would actually be a red flag. The result suggests that a simple 3-feature K-Means model can recover biologically plausible subtype structure, which is more useful than a null result.

On the T3DM hypothesis: The significant Spearman correlation ($\rho_s = 0.208$, $p = 5.29 \times 10^{-5}$) with Holm correction provides the first computational, open-data corroboration of the T3DM hypothesis using this specific dataset. The effect size is moderate ($\rho_s \approx 0.21$), which is consistent with the expected complex, polygenic, and lifestyle-mediated pathway between peripheral glycaemic control and central neural insulin signalling. The failure of the Kruskal-Wallis group test (H1) does not contradict this: a continuous-variable correlation (ρ_s) has greater statistical power than a three-group omnibus test in this small and unbalanced cohort.

Limitations:

- (i) The NCSU dataset lacks published size metadata; we report available sample statistics but cannot make population-level prevalence claims.

- (ii) The K-Means clustering lacks ground-truth type labels for external validation; the T1DM/T2DM assignment is therefore inferential.
- (iii) The Ohio dataset is cross-sectional in our analysis; longitudinal trajectory modelling would strengthen the T3DM causal argument.
- (iv) The Stacking Ensemble CV results were not fully converged within the computational budget and are reported as pending in Table 1.

7 Conclusion

We have presented a unified three-stage machine-learning framework for diabetes research that advances beyond binary detection to encompass subtype discrimination and cognitive-metabolic hypothesis testing. Our key empirical findings are:

- (1) SVM-RBF and Logistic Regression achieve the highest ROC-AUC of 0.825 ± 0.026 under stratified 5-fold CV, with SVM-RBF reaching recall = 0.741 on the held-out test set, the most clinically relevant metric for diabetes screening.
- (2) SHAP attribution robustly identifies Glucose, BMI, and Age as the three dominant predictive biomarkers, providing interpretable, clinically actionable insight alongside model predictions.
- (3) K-Means clustering ($k=2$, silhouette ≈ 0.116) of the diabetic sub-cohort recovers a biologically plausible T1DM/T2DM-aligned partition based solely on Glucose, Insulin, and Age.
- (4) A statistically significant positive Spearman correlation ($\rho_s = 0.208$, $p = 5.29 \times 10^{-5}$, Holm-corrected) between glycaemic control and cognitive function in the Ohio Longitudinal Dataset provides the first open-data computational corroboration of the T3DM hypothesis.

Future work will extend this framework with deep learning on longitudinal EHR sequences, external validation on population-scale cohorts, and a prospective clinical trial design for the T3DM component.

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